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Blind fastener

Abstract

A method includes inserting a machining tool through a hole of at least one workpiece that includes a first surface and a second surface opposite the first surface and extends from the first surface to the second surface. The hole forms an inner wall. The machining tool includes a shaft and a cutting tip proximate to one end of the shaft. The cutting tip includes a cutting portion. The method includes positioning the cutting portion of the cutting tip to abut an edge of the hole. The edge is located at a junction of the second surface of the one of the at least one workpiece and the inner wall. The method includes forming a surface modification at the edge with the cutting tip. The method includes removing the machining tool from the hole. The method includes installing a fastener within the hole. The fastener engages the surface modification.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional application relating to and claiming the benefit of commonly owned, co-pending U.S. patent application Ser. No. 17/542,036, filed Dec. 3, 2021, entitled “BLIND FASTENER,” the contents of which are incorporated by reference herein its entirety.

FIELD

(1) This disclosure relates generally to fasteners. More particularly, this disclosure relates to fasteners such as blind fasteners for securing a plurality of workpieces together.

BACKGROUND

(2) Blind fasteners are commonly used to secure a plurality of workpieces together when it is difficult or impossible to access a blind side of one of the workpieces. Blind fasteners generally include a deformable sleeve that, when deformed, engages a surface of the blind side of one of the workpieces to hold the workpieces together. Installation of fasteners tend to create high stress areas, particularly in composite structures.

SUMMARY

(3) In some embodiments, a method includes inserting a machining tool through a hole of at least one workpiece. In some embodiments, the at least one workpiece includes a first surface and a second surface opposite the first surface. In some embodiments, the hole extends from the first surface to the second surface. In some embodiments, the hole forms an inner wall. In some

embodiments, the machining tool includes a shaft, and a cutting tip located proximate to one end of the shaft. In some embodiments, the cutting tip includes a cutting portion. In some embodiments, the method includes positioning the cutting portion of the cutting tip to abut an edge of the hole of the one of the at least one workpiece. In some embodiments, the edge is located at a junction of the second surface of the one of the at least one workpiece and the inner wall. In some embodiments, the method includes forming a surface modification at the edge with the cutting tip. In some embodiments, the method includes removing the machining tool from the hole. In some embodiments, the method includes installing a fastener within the hole. In some embodiments, the fastener engages the surface modification.

(4) In other embodiments of the method, the at least one workpiece may additionally and/or alternatively include a plurality of workpieces, the plurality of workpieces being configured to be joined to one another, and holes in the plurality of workpieces being substantially aligned with one another.

(5) In other embodiments of the method, the method may additionally and/or alternatively include the step of forming the hole with the machining tool.

(6) In other embodiments of the method, the step of forming the hole may additionally and/or alternatively include drilling.

(7) In other embodiments of the method, the step of forming the hole and the step of inserting the machining tool into the hole may additionally and/or alternatively be performed concurrently.

(8) In other embodiments of the method, the cutting tip may additionally and/or alternatively be moveable between a retracted position in which the cutting tip is substantially positioned within the shaft, and an extended position in which the cutting tip extends outwardly from the shaft.

(9) In other embodiments of the method, the cutting tip may additionally and/or alternatively include a first ramped portion configured to contact the inner wall of the at least one workpiece to facilitate moving the cutting tip to its retracted position during the step of inserting the machining tool through the hole of the least one workpiece.

(10) In other embodiments of the method, the cutting tip may additionally and/or alternatively be in its extended position during the steps of positioning the cutting portion of the cutting tip and forming a surface modification.

(11) In other embodiments of the method, the cutting tip may additionally and/or alternatively include a second ramped portion configured to contact the inner wall to facilitate moving the cutting tip to its retracted position during the step of removing the machining tool from the hole.

(12) In other embodiments of the method, the step of forming the surface modification may additionally and/or alternatively include rotating the machining tool.

(13) In some embodiments, a machining tool includes a shaft; and a cutting tip located proximate to one end of the shaft. In some embodiments, the cutting tip includes a cutting portion. In some embodiments, the cutting tip includes a first ramped portion extending in a first direction relative to a longitudinal axis of the shaft. In some embodiments, the cutting tip includes a second ramped portion extending in a second direction relative to the longitudinal axis of the shaft.

(14) In other embodiments of the machining tool, the first direction may additionally and/or alternatively be oblique relative to the longitudinal axis and the second direction is oblique relative to the longitudinal axis.

(15) In other embodiments of the machining tool, the shaft may additionally and/or alternatively include a first opening, wherein the cutting portion is configured to be retractable into the shaft in a retracted position at the first opening and configured to be extendable from the shaft in an extended position at the first opening.

(16) In other embodiments of the machining tool, the machining tool may additionally and/or alternatively include a spring member located within the shaft, wherein the spring member is attached to the cutting tip, and wherein the spring member is configured to facilitate movement of the cutting tip to and from its retracted position to its extended position.

- (17) In other embodiments of the machining tool, the machining tool may additionally and/or alternatively include a pin disposed within the shaft and extending transversely to the longitudinal axis of the shaft.
- (18) In other embodiments of the machining tool, the spring member may additionally and/or alternatively be rotatable about the pin between the extended position and the retracted position.
- (19) In other embodiments of the machining tool, the cutting portion may additionally and/or alternatively include a concave cutting surface.
- (20) In other embodiments of the machining tool, the cutting tip may additionally and/or alternatively be composed of carbide.
- (21) In other embodiments of the machining tool, the cutting tip may additionally and/or alternatively be coated with a diamond coating.
- (22) In other embodiments of the machining tool, the machining tool may additionally and/or alternatively include a drill bit located at the one end of the shaft.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) References are made to the accompanying drawings that form a part of this disclosure and that illustrate embodiments in which the systems and methods described in this Specification can be practiced.
- (2) FIG. 1 illustrates a blind fastener shown in a pre-installed position, according to some embodiments.
- (3) FIG. 2 illustrates the blind fastener of FIG. 1 shown in an intermediate position during installation, according to some embodiments.
- (4) FIG. 3 illustrates the blind fastener of FIG. 1 shown in an installed position, according to some embodiments.
- (5) FIG. 4 is a photomicrograph of a portion of the fastener of FIG. 1 shown in the installed position of FIG. 3, according to some embodiments.
- (6) FIGS. 5A through 5C illustrate a machining tool, according to some embodiments.
- (7) FIGS. 6A through 6F illustrate a process of preparing a workpiece prior to installation of the blind fastener shown in FIG. 1, according to some embodiments.
- (8) Like reference numbers represent the same or similar parts throughout.

DETAILED DESCRIPTION

- (9) Referring to FIGS. 1 through 4, in some embodiments, a fastener **10** is a “threaded type” fastener. In some embodiments, the fastener **10** includes a core bolt **12** (e.g., a pin member), a sleeve **14**, and an insert **16**. In some embodiments, the fastener **10** is configured to secure a plurality of workpieces **18**, **20** to one another, and is configured to be installed within holes **22**, **24** formed within the workpieces **18**, **20**. In some embodiments, the holes **22**, **24** are substantially aligned with one another. In some embodiments, the holes **22**, **24** are aligned with one another. In some embodiments, the workpiece **20** includes a blind side **26** that may be difficult or impossible to access by a user with a fastener installation tool or otherwise. In some embodiments, a portion of each of the holes **22**, **24** has an inner diameter **28**. In some embodiments, the hole **22** includes a countersunk portion.
- (10) In some embodiments, the workpiece **18** is composed of a composite material. In some embodiments, the workpiece **18** is substantially composed of a composite material. In some embodiments, the workpiece **18** is partially composed of a composite material. In some embodiments, the workpiece **18** is composed of a metallic material. In some embodiments, the workpiece **18** is substantially composed of a metallic material. In some embodiments, the workpiece **18** is partially composed of a metallic material. In some embodiments, the workpiece **18**

is composed of aluminum.

(11) In some embodiments, the workpiece **20** is composed of a composite material. In some embodiments, the workpiece **20** is substantially composed of a composite material. In some embodiments, the workpiece **20** is partially composed of a composite material. In some embodiments, the workpiece **20** is composed of a metallic material. In some embodiments, the workpiece **20** is substantially composed of a metallic material. In some embodiments, the workpiece **20** is partially composed of a metallic material. In some embodiments, the workpiece **20** is composed of aluminum.

(12) In some embodiments, the core bolt **12** includes an elongated shank portion **30** extending along a longitudinal axis between a first end **32** and a second end **34** opposite the first end **32**. In some embodiments, the core bolt **12** terminates with a head **36** at the first end **32**. In an embodiment, the head **36** is an enlarged head. In some embodiments, the head **36** may include other shapes and sizes. In some embodiments, a fastener head **38**, which is configured to be gripped by an installation tool, extends from the head **36** in a direction away from the elongated shank portion **30**. In some embodiments, the fastener head **38** includes a plurality of splines **40** that are configured to be gripped by a correspondingly shaped fastener installation tool. In some embodiments, the elongated shank portion **30** includes a substantially smooth cylindrical portion **42** adjacent the first end **32**, a threaded portion **44** adjacent the second end **34**, and a thread run-out **46** (i.e., a step) intermediate the smooth cylindrical portion **42** and the threaded portion **44**. In some embodiments, the smooth cylindrical portion **42** has an outer diameter **48**. In some embodiments, the threaded portion **44** has a major diameter **50** that is less than the outer diameter **48** of the smooth cylindrical portion **42**. In some embodiments, the threaded portion **44** includes a minor diameter **52**.

(13) Still referring to FIGS. **1** through **3**, in some embodiments, the sleeve **14** includes a tubular portion **60** having a first end **62** and a second end **64** opposite the first end **62**. In some embodiments, the sleeve **14** terminates with a flanged head **66** at the first end **62**. In some embodiments, a pocket **68** is formed within the flanged head **66**. In some embodiments, the pocket **68** is sized and shaped to receive the head **36** of the core bolt **12**. In some embodiments, the tubular portion **60** includes an outer diameter **70** that is sized and shaped to enable installation of the sleeve **14** within the holes **22**, **24** of the workpieces **18**, **20**. In some embodiments, the tubular portion **60** includes a first region **72** adjacent the first end **62**, a second region **74** intermediate the first end **62** and the second end **64**, and a third region **76** adjacent the second end **64**. In some embodiments, the first region **72** includes an inner diameter **78** that is sized and shaped to enable installation of the core bolt **12** within the sleeve **14**. In some embodiments, the second region **74** includes an inner diameter **80** that is less than the inner diameter **78** of the first region **72** and greater than the major diameter **50** of the threaded portion **44** of the core bolt **12**. In some embodiments, a step **82** is formed within the sleeve **14** and provides a stepped transition between the inner diameter **78** of the first region **72** and the inner diameter **80** of the second region **74**. In some embodiments, the third region **76** includes internal threads **84** that are complementary to and configured to engage threadedly the threaded portion **44** of the core bolt **12**. In some embodiments, the sleeve **14** includes a narrowed tip **86** adjacent the second end **64**. In some embodiments, the narrowed tip **86** has an outer diameter **88** that is less than the outer diameter **70** of the tubular portion **60**. In some embodiments, the sleeve **14** includes a band annealed portion positioned so as to be proximate the insert **16** when the fastener **10** is in a pre-installation position. In some embodiments, band annealing may be performed using a laser. In some embodiments, band annealing may be performed using a radio frequency induction coil. In some embodiments, band annealing may be performed using another suitable means known in the art.

(14) In some embodiments, the insert **16** has a substantially tubular portion **90** extending from a first end **92** to a second end **94** opposite the first end **92**. The tubular portion **90** has an outer diameter **96** that is sized and shaped to enable the installation of the insert **16** within the sleeve **14**.

In some embodiments, the outer diameter **96** of the insert **16** is substantially equal to the outer diameter **48** of the smooth cylindrical portion **42** of the core bolt **12**. In some embodiments, the tubular portion **90** has internal threads **98** that are complementary to the threaded portion **44** of the core bolt **12**. In some embodiments, the tubular portion **90** has a length **100** that is selected such that, when the fastener **10** is in a pre-installation position (see FIG. 1), the first end **92** abuts the thread run-out **46** of the core bolt **12** and the second end **94** abuts the step **82** of the sleeve **14**. In some embodiments, the insert **16** is formed from a material that has lubricating properties. In some embodiments, the insert **16** includes copper. In other embodiments, the insert **16** may include other materials (e.g., other metals, plastics, etc.) that have frictional properties suitable to prevent thread galling and that are sufficiently malleable to aid in bulb formation, as will be described in further detail hereinafter.

(15) In some embodiments, the core bolt **12** is composed of steel. In some embodiments, the core bolt **12** is composed of stainless steel. In some embodiments, the core bolt **12** is composed of 300 series stainless steel. In some embodiments, the core bolt **12** is composed of alloy steel. In some embodiments, the core bolt **12** is composed of a corrosion resistant material. In some embodiments, the core bolt **12** is composed of titanium alloy. In some embodiments, the core bolt **12** is composed of nickel alloy. In some embodiments, the sleeve **14** is composed of copper. In some embodiments, the sleeve **14** is composed of brass. In some embodiments, the sleeve **14** is composed of aluminum alloy. In some embodiments, the sleeve **14** is composed of 6066 aluminum alloy. In some embodiments, the sleeve **14** is composed of stainless steel. In some embodiments, the sleeve **14** is composed of **300** series stainless steel.

(16) In some embodiments, the insert **16** is composed of metal. In some embodiments, the insert **16** is composed of copper. In some embodiments, the insert **16** is formed from a material that has lubricating properties. In some embodiments, the insert **16** includes other materials (e.g., other metals, plastics, etc.) that have frictional properties suitable to prevent thread galling and that are sufficiently malleable to aid in bulb formation.

(17) Referring now to FIG. 1, in some embodiments, a pre-installation position of the fastener **10** is shown. In some embodiments, the internal threads **98** of the insert **16** receive and engage threadedly the threaded portion **44** of the core bolt **12** such that the first end **92** of the insert **16** abuts the thread run-out **46** of the core bolt **12**. In some embodiments, the internal threads **84** of the sleeve **14** receive and engage threadedly the threaded portion **44** of the core bolt **12** such that the step **82** abuts the second end **94** of the insert **16**, thereby trapping the insert **16** between the thread run-out **46** of the core bolt **12** and the step **82** of the sleeve **14**. In some embodiments, the head **36** of the core bolt **12** is not seated within the pocket **68** of the sleeve **14**, and is positioned such that it will seat within the pocket **68** of the sleeve **14** after some rotations (e.g., two to three rotations) of the core bolt **12** with respect to the sleeve **14**.

(18) Referring to FIGS. 1 through 4, in some embodiments, the workpiece **20** includes a surface modification **110** at a surface that is to be gripped by the fastener **10**. In some embodiments, the surface modification **110** includes a rounding of the workpiece **20**. That is, in some embodiments, prior to installation of the fastener **10**, the workpiece **20** is prepared by rounding a corner to form the surface modification **110**. In some embodiments, the surface modification **110** includes a radius **112**. In some embodiments, the radius **112** is selected based on a diameter of the holes **22**, **24**. In some embodiments, the surface modification **110** results in an improved extrusion of the insert **16** during installation of the fastener **10**. In some embodiments, the formation of the surface modification **110**, which can be referred to as a convex portion, provides an improved fatigue resistance by reducing a sharp contact point between the workpiece **20** and the fastener **10**. In some embodiments, a bulb **102** of the sleeve **14** contacts the surface modification **110**. In some embodiments, the bulb **102** includes a radius that is sized and shaped to conform to the surface modification **110**. In some embodiments, as the diameter of the holes **22**, **24** increases above $\frac{3}{8}$ ", an increase in the improvement of the fatigue resistance is also increased. In some embodiments, the

extrusion of the insert **16** between the sleeve **14** and the core bolt **12** creates a compressive stress, which serves as a countermeasure to the initiation of a fatigue crack.

(19) Referring now to FIG. 2, an intermediate position of the fastener **10** during installation is shown. During installation, a torque is applied to the fastener head **38** by an installation tool that is configured to grip the splines **40**. In some embodiments, as the core bolt **12** rotates with respect to the sleeve **14**, the insert **16** is compressed between the thread run-out **46** of the core bolt **12** and the step **82** of the sleeve **14**. In some embodiments, in response to such compression, the insert **16** can only deform in an outward radial direction (i.e., away from the core bolt **12**). In some embodiments, deformation of the insert **16** causes corresponding deformation of the sleeve **14** in an outward direction (i.e., away from the core bolt **12**). In some embodiments, before the head **36** of the core bolt **12** is seated within the pocket **68** of the sleeve **14**, no compressive force is exerted by the core bolt **12** on the sleeve **14**. In some embodiments, with continued rotation of the core bolt **12** with respect to the sleeve **14**, the head **36** of the core bolt **12** advances to a point such that it is seated within the pocket **68** of the sleeve **14**.

(20) Continuing to refer to FIG. 2, in some embodiments, once the head **36** of the core bolt **12** is seated within the pocket **68** of the sleeve **14**, continued rotation of the core bolt **12** with respect to the sleeve **14** continues to compress the insert **16**. In some embodiments, the insert **16** is constrained between the thread run-out **46** of the core bolt **12** and the step **82** of the sleeve **14**, and as a result, it can only deform outwardly (i.e., away from the core bolt **12**), and the sleeve **14** deforms correspondingly. In some embodiments, such deformation of the insert **16** and the sleeve **14** causes the formation of the bulb **102**.

(21) FIG. 3 illustrates the fastener **10** after the bulbing action of the insert **16** has been completed, in accordance with some embodiments. In some embodiments, a bulb **102** is formed by the insert **16** and the sleeve **14**. In some embodiments, the bulb **102** has a diameter that is at least 1.4 times the outer diameter **70** of the tubular portion **60** of the sleeve **14**. In some embodiments, the workpieces **18**, **20** are secured between the bulb **102**, on the blind side **26** of the workpiece **20**, and the flanged head **66** of the sleeve **14**, on the side of the workpiece **18** that is accessible by an installation tool. As shown in FIG. 3, in some embodiments, the fastener head **38** may be sheared off from the core bolt **12** to produce a fastener **10** that is substantially flush with a surface of the workpiece **18** when the fastener **10** is in its installed position.

(22) Referring to FIG. 1, in some embodiments, the fastener **10** is configured to form the bulb **102** as described above when installed in workpieces **18**, **20** having a total combined thickness of between a minimum grip **106** and a maximum grip **108**. Referring to FIG. 4, in some embodiments, the insert **16** is swaged outwardly to form the bulb **102**.

(23) Referring to FIGS. 5A through 5C, in some embodiments, a machining tool **150** includes a shaft **152** with a cutting tip **154** at one end thereof. In some embodiments, the cutting tip **154** is retractable into the shaft **152** through a first opening **156** in the shaft **152**. In some embodiments, the first opening **156** enables the cutting tip **154** to retract into the shaft **152** so that the machining tool **150** can be inserted into and removed from the holes **22**, **24**. In some embodiments, the machining tool **150** is configured to prepare the workpiece **20** for the fastener **10**. In some embodiments, the machining tool **150** is configured to prepare the surface modification **110** of the workpiece **20**. In some embodiments, the machining tool **150** is a standalone tool. In some embodiments, the machining tool **150** can be combined with, for example, a drill bit or the like that forms the holes **22**, **24** in the workpieces **18**, **20**. In such embodiments, the drill bit can be used to create the holes **22**, **24** in the workpieces **18**, **20**, and after which, once inserted far enough, the cutting tip **154** can protrude from the shaft **152** and be rotated to remove a portion of material of the workpiece **20** to form the surface modification **110**. In some embodiments, the surface modification **110** is convex. In some embodiments, the machining tool **150** can be a separate tool that is inserted after the drill bit is removed.

(24) In some embodiments, the cutting tip **154** is composed of a hardened material. In some

embodiments, the cutting tip **154** is composed of a material that is harder than the workpiece **20**. In some embodiments, the cutting tip **154** is capable of cutting away part of the material of the workpiece **20**. In some embodiments, the cutting tip **154** is composed of carbide. In some embodiments, the cutting tip **154** is coated with a diamond coating.

(25) In some embodiments, the cutting tip **154** includes a first ramped portion **158**, a second ramped portion **160**, and a cutting portion **162**. In some embodiments, the first ramped portion **158** extends obliquely relative to a longitudinal axis of the shaft **152**. In some embodiments, the second ramped portion **160** extends obliquely relative to a longitudinal axis of the shaft **152**. In some embodiments, the first ramped portion **158** extends in a first direction, while the second ramped portion **160** extends in a second direction. In some embodiments, the first ramped portion **158** and the second ramped portion **160** are configured to enable the cutting tip **154** to extend or retract from the shaft **152** based on contact with the workpiece **18** or the workpiece **20**. In some embodiments, the cutting portion **162** has a concave cutting surface. In some embodiments, the cutting portion **162** includes an inner radius. In some embodiments, when the cutting portion **162** contacts the workpiece **20**, the shaft **152** is rotated to remove a portion of the workpiece **20**. In some embodiments, the cutting portion **162** is sized and shaped to form a corresponding size and shape of the radius **112** of the surface modification **110**.

(26) In some embodiments, the shaft **152** includes a second opening **164**. In some embodiments, the second opening **164** is located proximate to the first opening **156**. In some embodiments, the second opening **164** is axially formed within the shaft **152**. In some embodiments, the second opening **164** enables the cutting tip **154** to be removed from the machining tool **150**. In some embodiments, the cutting tip **154** is replaceable.

(27) In some embodiments, the shaft **152** includes a third opening **166** aligned with the first opening **156**. In some embodiments, the third opening **166** is configured to enable the cutting tip **154** to retract into the shaft **152**. In some embodiments, the shaft includes a spring member **168**, one end of which is attached to a pin **170**. In some embodiments, the pin **170** extends transversely to a longitudinal axis of the shaft **152**. In some embodiments, the spring member **168** is rotatable about the pin **170** to enable the cutting tip **154** to move from a first retracted position, in which the cutting tip **154** is retracted within the shaft **152**, to a second position, in which the cutting tip **154** extends outwardly from the first opening **156** of the shaft **152**. In some embodiments, the cutting tip **154** is fixed on the shaft **152**.

(28) Referring to FIGS. **6A** through **6F**, in some embodiments, a process of preparing the workpiece **20** prior to installation of a fastener **10** includes preparation of the workpiece **20** with the machining tool **150**. In some embodiments, the workpiece **20** is prepared using the machining tool **150** when not assembled with workpiece **18**. In FIGS. **6A** through **6F**, the machining tool **150** is shown in sectional view.

(29) In some embodiments, prior to insertion of the machining tool **150** into the holes **22**, **24** (see FIG. **6A**), the cutting tip **154** extends from the shaft **152**. In some embodiments, when the first ramped portion **158** contacts a wall of the holes **22**, **24** (FIG. **6B**), the contact forces the cutting tip **154** to be retracted into the shaft **152**, enabling the machining tool **150** to be inserted into the holes **22**, **24**. In some embodiments, once the machining tool **150** is inserted through the holes **22**, **24** (FIG. **6C**), the cutting tip **154** protrudes from the shaft **152**. In some embodiments, the machining tool **150** is retracted in the holes **22**, **24** until the cutting portion **162** abuts an edge of the blind side surface of the workpiece **20** (FIG. **6D**) and the holes **22**, **24**. In some embodiments, with the cutting portion **162** abutting the workpiece **20**, the machining tool **150** is rotated, cutting away the surface of the workpiece **20** and forming the surface modification **110** in the workpiece **20** (FIG. **6D**). In some embodiments, after the surface modification **110** is formed, the machining tool **150** is retracted from the holes **22**, **24**. In some embodiments, the second ramped portion **160** contacts the wall of the holes **22**, **24** and forces the cutting tip **154** to be retracted into the shaft **152** (FIG. **6E**). In some embodiments, the machining tool **150** is removed from the holes **22**, **24** and the

workpieces **18, 20** (FIG. 6F). In some embodiments, the workpieces **18, 20** are disassembled from one another, cleaned to eliminate any loose material from the cutting process, and reassembled with one another. In some embodiments, the workpieces **18, 20** remain assembled during the cutting and cleaning processes.

(30) The terminology used herein is intended to describe embodiments and is not intended to be limiting. The terms “a,” “an,” and “the” include the plural forms as well, unless clearly indicated otherwise. The terms “comprises” and/or “comprising,” when used in this Specification, specify the presence of the stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, and/or components.

(31) It is to be understood that changes may be made in detail, especially in matters of the construction materials employed and the shape, size, and arrangement of parts without departing from the scope of the present disclosure. This Specification and the embodiments described are examples, with the true scope and spirit of the disclosure being indicated by the claims that follow.

Claims

1. A machining tool, comprising: a shaft having a longitudinal axis; and a cutting tip located proximate to one end of the shaft, wherein the cutting tip includes a first side, a second side opposite the first side, and a width extending from the first side to the second side, a first edge, a second edge opposite the first edge, a cutting portion, wherein the cutting portion includes a concave cutting surface extending towards the second edge from the first edge to a first location between the first edge and the second edge, wherein the concave cutting surface extends across an entirety of the width of the cutting tip, wherein the concave cutting surface includes an inner radius, wherein the inner radius of the concave cutting surface is sized and shaped to form a surface modification having an outer radius corresponding to the inner radius of the concave cutting surface, a first ramped portion extending obliquely in a first direction relative to a longitudinal axis of the shaft, wherein the first ramped portion extends from the second edge to a second location between the first edge and the second edge, and a second ramped portion extending uniformly obliquely in a second direction relative to the longitudinal axis of the shaft, wherein the second direction is different than the first direction, wherein the second ramped portion extends from the first location to the first ramped portion at the second location, wherein the shaft comprises a first opening, and wherein the cutting portion is configured to be retractable into the shaft in a retracted position at the first opening and configured to be extendable from the shaft in an extended position at the first opening.
 2. The machining tool of claim 1, further comprising a drill bit located at the one end of the shaft.
 3. The machining tool of claim 1, wherein the cutting tip is composed of carbide.
 4. The machining tool of claim 3, wherein the cutting tip is coated with a diamond coating.
 5. The machining tool of claim 1, wherein the machining tool includes a spring member located within the shaft, wherein the spring member is attached to the cutting tip, and wherein the spring member is configured to facilitate movement of the cutting tip to and from its retracted position to its extended position.
 6. The machining tool of claim 5, further comprising a pin disposed within the shaft and extending transversely to the longitudinal axis of the shaft.
 7. The machining tool of claim 6, wherein the spring member is rotatable about the pin between the extended position and the retracted position.
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